PADHAKU
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CHAPTER 4

Exploring Algebraic Identities

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ 4.1 Introduction

? **Add the smallest and largest of the consecutive squares 9, 16, 25 and subtract twice the middle square.**

✔ ANSWER

$(9 + 25) - (2 \times 16)$

$= 34 - 32$

$= \mathbf{2}$

Think and Reflect **Find other patterns like this one. For example, consider 4 consecutive squares and see if you can find a pattern.**

💡 SAMPLE ANSWER

Take 4 consecutive squares, e.g. 1, 4, 9, 16.

(smallest + largest) − (sum of the two middle squares) = (1 + 16) − (4 + 9) = 17 − 13 = 4

Check: 4, 9, 16, 25 → (4 + 25) − (9 + 16) = 29 − 25 = 4

Proof: $n^2 + (n + 3)^2 - (n + 1)^2 - (n + 2)^2$

$= n^2 + n^2 + 6n + 9 - n^2 - 2n - 1 - n^2 - 4n - 4 = 4$

The answer is always 4.

🤔 Think and Reflect · 4.2 Visualising Identities

1 **What can you say about a and b if $(a + b)^2 < a^2 + b^2$?**

✔ ANSWER

$(a + b)^2 = a^2 + b^2 + 2ab$, so the difference is decided by 2ab.

$(a + b)^2 < a^2 + b^2$ when $2ab < 0$, i.e. **ab < 0: a and b have opposite signs** (one positive, one negative).

2 **What can you say about a and b if $(a + b)^2 > a^2 + b^2$?**

✔ ANSWER

$(a + b)^2 > a^2 + b^2$ when $2ab > 0$, i.e. **ab > 0: a and b have the same sign** (both positive or both negative).

3 **When will $(a + b)^2$ be equal to $a^2 + b^2$?**

✔ ANSWER

When $2ab = 0$, i.e. **a = 0 or b = 0**.

📝 Exercise Set 4.1

1 (i) **Expand $(7x + 4y)^2$**

✔ ANSWER

$(7x)^2 + 2(7x)(4y) + (4y)^2$

= **$49x^2 + 56xy + 16y^2$**

1 (ii) **Expand $(7/5\ x + 3/2\ y)^2$**

✔ ANSWER

$(7/5\ x)^2 + 2(7/5\ x)(3/2\ y) + (3/2\ y)^2$

= **$49/25\ x^2 + 21/5\ xy + 9/4\ y^2$**

1 (iii) **Expand $(2.5p + 1.5q)^2$**

✔ ANSWER

$(2.5p)^2 + 2(2.5p)(1.5q) + (1.5q)^2$

= **$6.25p^2 + 7.5pq + 2.25q^2$**

1 (iv) **Expand $(3/4\ s + 8t)^2$**

✔ ANSWER

$(3/4\ s)^2 + 2(3/4\ s)(8t) + (8t)^2$

= **$9/16\ s^2 + 12st + 64t^2$**

1 (v) **Expand $(x + 1/(2y))^2$**

✔ ANSWER

$x^2 + 2(x)(1/(2y)) + (1/(2y))^2$

= **$x^2 + x/y + 1/(4y^2)$**

1 (vi) **Expand $(1/x + 1/y)^2$**

✔ ANSWER

$(1/x)^2 + 2(1/x)(1/y) + (1/y)^2$

$= \mathbf{1/x^2 + 2/(xy) + 1/y^2}$

2 (i) **Find $(64)^2$**

✔ ANSWER

$(60 + 4)^2 = 60^2 + 2 \times 60 \times 4 + 4^2$

$= 3600 + 480 + 16 = \mathbf{4096}$

2 (ii) **Find $(105)^2$**

✔ ANSWER

$(100 + 5)^2 = 100^2 + 2 \times 100 \times 5 + 5^2$

$= 10000 + 1000 + 25 = \mathbf{11025}$

2 (iii) **Find $(205)^2$**

✔ ANSWER

$(200 + 5)^2 = 200^2 + 2 \times 200 \times 5 + 5^2$

$= 40000 + 2000 + 25 = \mathbf{42025}$

🤔 Think and Reflect · 4.3 Factorisation of Algebraic Expressions Using Identities

1 **What if we replace b by −b in $(a + b)^2 = a^2 + 2ab + b^2$?**

✔ ANSWER

$(a + (-b))^2 = a^2 + 2a(-b) + (-b)^2$

$(a - b)^2 = a^2 - 2ab + b^2$, which is also an identity.

📝 Exercise Set 4.2

1 (i) **Factor completely: $9x^2 + 24xy + 16y^2$**

✔ ANSWER

$= (3x)^2 + 2(3x)(4y) + (4y)^2$

$= \mathbf{(3x + 4y)^2}$

1 (ii) **Factor completely: $4s^2 + 20st + 25t^2$**

✔ ANSWER

$= (2s)^2 + 2(2s)(5t) + (5t)^2$

$= \mathbf{(2s + 5t)^2}$

1 (iii) **Factor completely: $49x^2 + 28xy + 4y^2$**

✔ ANSWER

$= (7x)^2 + 2(7x)(2y) + (2y)^2$

$= \mathbf{(7x + 2y)^2}$

1 (iv) **Factor completely: $64p^2 + 32/3\ pq + 4/9\ q^2$**

✔ ANSWER

$= (8p)^2 + 2(8p)(2/3\ q) + (2/3\ q)^2$

$= \mathbf{(8p + 2/3\ q)^2}$

1 (v)* **Factor completely: $3a^2 + 4ab + 4/3\ b^2$**

✔ ANSWER

Take 1/3 out as a common factor: = 1/3 $(9a^2 + 12ab + 4b^2)$

= 1/3 $[(3a)^2 + 2(3a)(2b) + (2b)^2]$

= **$1/3\ (3a + 2b)^2$**

(Same as $3(a + 2/3\ b)^2$.)

1 (vi)* **Factor completely: $9/5\ s^2 + 6sv + 5v^2$**

✔ ANSWER

Take 1/5 out: = 1/5 $(9s^2 + 30sv + 25v^2)$

= 1/5 $[(3s)^2 + 2(3s)(5v) + (5v)^2]$

= **$1/5\ (3s + 5v)^2$**

2 (i) **Find $(79)^2$ using $(a - b)^2 = a^2 - 2ab + b^2$**

✔ ANSWER

$(80 - 1)^2 = 6400 - 160 + 1 =$ **6241**

2 (ii) **Find $(193)^2$**

✔ ANSWER

$(200 - 7)^2 = 40000 - 2800 + 49 =$ **37249**

2 (iii) **Find $(299)^2$**

✔ ANSWER

$(300 - 1)^2 = 90000 - 600 + 1 = \mathbf{89401}$

🤔 Think and Reflect · 4.4 More Identities

1 **Label the squares and rectangles in Fig. 4.4 so that it represents $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.**

✔ ANSWER

The big square has side a + b + c, split into 3 × 3 = 9 parts.

	a	b	c
a	a^2	ab	ac
b	ab	b^2	bc
c	ac	bc	c^2

Total = $a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ = $\mathbf{(a + b + c)^2}$.

📝 Exercise Set 4.3

1 (i) **Find 117^2**

✔ ANSWER

$(100 + 10 + 7)^2 = 100^2 + 10^2 + 7^2 + 2(100)(10) + 2(10)(7) + 2(100)(7)$

$= 10000 + 100 + 49 + 2000 + 140 + 1400 =$ **13689**

1 (ii) **Find 78^2**

✔ ANSWER

$(80 - 2)^2 = 6400 - 320 + 4 =$ **6084**

1 (iii) **Find 198^2**

✔ ANSWER

$(200 - 2)^2 = 40000 - 800 + 4 =$ **39204**

1 (iv) **Find 214^2**

✔ ANSWER

$(200 + 10 + 4)^2 = 40000 + 100 + 16 + 4000 + 80 + 1600 =$ **45796**

1 (v) **Find 1104^2**

✔ ANSWER

$(1100 + 4)^2 = 1210000 + 2 \times 1100 \times 4 + 16$

$= 1210000 + 8800 + 16 =$ **1218816**

1 (vi) **Find 1120^2**

✔ ANSWER

$(1000 + 100 + 20)^2 = 1000000 + 10000 + 400 + 200000 + 4000 + 40000$

$=$ **1254400**

$(a + b)^2$ suits 1104 and 78/198 suit $(a - b)^2$; $(a + b + c)^2$ suits 117, 214, 1120.

2 (i) **Factor: $16y^2 - 24y + 9$**

✔ ANSWER

$= (4y)^2 - 2(4y)(3) + 3^2$

$= \mathbf{(4y - 3)^2}$

2 (ii) **Factor: $9/4\ s^2 + 6st + 4t^2$**

✔ ANSWER

$= (3/2\ s)^2 + 2(3/2\ s)(2t) + (2t)^2$

$= \mathbf{(3/2\ s + 2t)^2}$

2 (iii) **Factor: $m^2/9 + mk/3 + k^2/4 + 3nk + 2mn + 9n^2$**

✔ ANSWER

Squares: $(m/3)^2$, $(k/2)^2$, $(3n)^2$

$2(m/3)(k/2) = mk/3$, $2(k/2)(3n) = 3nk$, $2(3n)(m/3) = 2mn$

$= \mathbf{(m/3 + k/2 + 3n)^2}$

2 (iv) **Factor: $p^2/16 - 2 + 16/p^2$**

✔ ANSWER

$= (p/4)^2 - 2(p/4)(4/p) + (4/p)^2$

$= \mathbf{(p/4 - 4/p)^2}$

2 (v) **Factor: $9a^2 + 4b^2 + c^2 - 12ab + 6ac - 4bc$**

✔ ANSWER

Squares: $(3a)^2$, $(2b)^2$, c^2. The signs show b has the minus sign.

$2(3a)(-2b) = -12ab$, $2(3a)(c) = 6ac$, $2(-2b)(c) = -4bc$

$= \mathbf{(3a - 2b + c)^2}$

3 (i) **Expand $(p + 3q + 7r)^2$**

✔ ANSWER

$p^2 + (3q)^2 + (7r)^2 + 2(p)(3q) + 2(3q)(7r) + 2(7r)(p)$

$= \mathbf{p^2 + 9q^2 + 49r^2 + 6pq + 42qr + 14pr}$

3 (ii) **Expand $(3x - 2y + 4z)^2$**

✔ ANSWER

$(3x)^2 + (-2y)^2 + (4z)^2 + 2(3x)(-2y) + 2(-2y)(4z) + 2(4z)(3x)$

$= \mathbf{9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24xz}$

4 **Is this an identity? $(a + b - c)^2 + (a - b + c)^2 + (a - b - c)^2 = 2a^2 + 2b^2 + 2c^2$**

✔ ANSWER

$(a + b - c)^2 = a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$

$(a - b + c)^2 = a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$

$(a - b - c)^2 = a^2 + b^2 + c^2 - 2ab + 2bc - 2ca$

Sum = $3a^2 + 3b^2 + 3c^2 - 2ab - 2bc - 2ca$

This is not equal to $2a^2 + 2b^2 + 2c^2$ for all values. Example: a = 1, b = 0, c = 0 gives LHS = 3, RHS = 2.

No, it is not an identity.

4.4 More Identities · Fig. 4.5 and Think and Reflect

? **Look at Fig. 4.5. Justify the identity $a^2 = (a + b)(a - b) + b^2$ for yourself.**

✔ ANSWER

Start with a square of side a (area a^2).

Cut a strip of width b from the bottom. It is a × b. Cut it into a (a − b) × b piece and a b × b square.

Move the (a − b) × b piece to the right side. Now we have a rectangle of sides (a + b) and (a − b).

The small b × b square is left over.

So $\mathbf{a^2 = (a + b)(a - b) + b^2}$.

1 **Evaluate using a suitable identity: (i) 35^2 (ii) 65^2 (iii) 85^2 (iv) 105^2. Do you observe any interesting pattern?**

✔ ANSWER

Use $a^2 = (a + 5)(a - 5) + 5^2$:

(i) $35^2 = 40 \times 30 + 25 =$ **1225**

(ii) $65^2 = 70 \times 60 + 25 =$ **4225**

(iii) $85^2 = 90 \times 80 + 25 =$ **7225**

(iv) $105^2 = 110 \times 100 + 25 =$ **11025**

Pattern: for a number ending in 5, multiply the number before 5 by the next number and write 25 after it (e.g. 6 × 7 = 42 → 4225).

2 **Observe the two rows of figures (Fig. 4.6). They represent an algebraic identity. Try to identify it.**

✔ ANSWER

Row 1: squares of sides a + b + c, a + b − c, a − b + c, a − b − c.

Row 2: squares of sides 2a, 2b, 2c.

$(a + b + c)^2 + (a + b - c)^2 + (a - b + c)^2 + (a - b - c)^2 = (2a)^2 + (2b)^2 + (2c)^2$

Check: the four squares give $4a^2 + 4b^2 + 4c^2$; all the ab, bc, ca terms cancel (two + and two −).

🤔 Think and Reflect · 4.5 Factorisation Using Algebra Tiles

1 **Suppose 7x is split as 2x + 5x; can a similar rectangular arrangement be formed? Consider other possibilities and check.**

✔ ANSWER

No. With 2 x-tiles on one side and 5 on the other, the unit tiles must fill a 2 by 5 array = 10 tiles, but we have 12.

Other splits: 1x + 6x needs 6 tiles, 2x + 5x needs 10. Only **3x + 4x** gives 3 × 4 = 12.

So only 3x + 4x forms a rectangle: (x + 3)(x + 4).

2 **Figure out the product of x + 2 and x + 3 using algebra tiles.**

✔ ANSWER

One x^2-tile, 2 x-tiles on one side, 3 x-tiles on the other, and 2 × 3 = 6 unit tiles in the corner.

$(x + 2)(x + 3) = x^2 + 5x + 6$

3 **Lay out algebra tiles for $x^2 + 11x + 30$ so that you see its factors.**

✔ ANSWER

Split 11x as 5x + 6x (because 5 × 6 = 30).

Place the x^2-tile, 5 x-tiles on the right, 6 x-tiles below, and 30 unit tiles in a 5 by 6 array.

The rectangle has sides x + 5 and x + 6: **$x^2 + 11x + 30 = (x + 5)(x + 6)$**

4 **$(x + 3)(x + 4) = x^2 + 7x + 12$ and $(x + 6)(x + 7) = x^2 + 13x + 42$. Generalise the pattern for $(x + a)(x + b)$.**

✔ ANSWER

Coefficient of x = sum of the numbers; constant = product.

(x + a)(x + b) = x² + (a + b)x + ab

5 **Fill in the blanks: (px + a)(qx + b) = (___)x² + (___)x + ___. Verify using the distributive property.**

✔ ANSWER

(px + a)(qx + b) = px(qx) + px(b) + a(qx) + ab

= **(pq)x² + (pb + aq)x + ab**

Example (Fig. 4.8): (2x + 3)(3x + 1) = 6x² + 2x + 9x + 3 = 6x² + 11x + 3.

📝 Exercise Set 4.4

1 (i) **s² − 11s + 24 = (____)(____)**

✔ ANSWER

Need a + b = −11, ab = 24 → −3 and −8.

s² − 11s + 24 = (s − 3)(s − 8)

1 (ii) **(____)(x + 1) = 3x² − 4x − 7**

✔ ANSWER

Try (3x − 7)(x + 1) = 3x² + 3x − 7x − 7 = 3x² − 4x − 7 ✓

(3x − 7)(x + 1)

1 (iii) **10x² − 11x − 6 = (2x − ___)(___ + 2)**

✔ ANSWER

$(2x - 3)(5x + 2) = 10x^2 + 4x - 15x - 6 = 10x^2 - 11x - 6$ ✓

$(2x - 3)(5x + 2)$

1 (iv) **$6x^2 + 7x + 2 = (____)(____)$**

✔ ANSWER

$(2x + 1)(3x + 2) = 6x^2 + 4x + 3x + 2 = 6x^2 + 7x + 2$ ✓

$(2x + 1)(3x + 2)$

2 (i) **$(41)^2$**

✔ ANSWER

$(40 + 1)^2 = 1600 + 80 + 1 =$ **1681**

2 (ii) **$(27)^2$**

✔ ANSWER

$(30 - 3)^2 = 900 - 180 + 9 =$ **729**

2 (iii) **23×17**

✔ ANSWER

$(20 + 3)(20 - 3) = 20^2 - 3^2 = 400 - 9 =$ **391**

2 (iv) **$(135)^2$**

✔ ANSWER

$(130 + 5)^2 = 16900 + 1300 + 25 =$ **18225**

2 (v) **$(97)^2$**

✔ ANSWER

$(100 - 3)^2 = 10000 - 600 + 9 =$ **9409**

2 (vi) **18×29**

✔ ANSWER

Use $(x + a)(x + b) = x^2 + (a + b)x + ab$ with $x = 20$, $a = -2$, $b = 9$:

$= 400 + 7 \times 20 + (-18) = 400 + 140 - 18 =$ **522**

2 (vii) **34×43**

✔ ANSWER

$x = 40$, $a = -6$, $b = 3$:

$= 1600 + (-3) \times 40 + (-18) = 1600 - 120 - 18 =$ **1462**

2 (viii) **$(205)^2$**

✔ ANSWER

$(200 + 5)^2 = 40000 + 2000 + 25 =$ **42025**

3 (i) **Factor: $9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$**

✔ ANSWER

Squares: $(3a)^2$, b^2, $(2c)^2$. $2(3a)(-b) = -6ab$, $2(3a)(2c) = 12ac$, $2(-b)(2c) = -4bc$

= **$(3a - b + 2c)^2$**

3 (ii) **Factor: $16s^2 + 25t^2 - 40st$**

✔ ANSWER

= $(4s)^2 - 2(4s)(5t) + (5t)^2$ = **$(4s - 5t)^2$**

3 (iii) **Factor: $r^2 - r - 42$**

✔ ANSWER

Need $a + b = -1$, $ab = -42$ → -7 and 6.

= **$(r - 7)(r + 6)$**

3 (iv) **Factor: $49g^2 + 14gh + h^2$**

✔ ANSWER

= $(7g)^2 + 2(7g)(h) + h^2$ = **$(7g + h)^2$**

3 (v) **Factor: $64u^2 + 121v^2 + 4w^2 - 176uv - 32uw + 44vw$**

✔ ANSWER

Squares: $(8u)^2$, $(11v)^2$, $(2w)^2$.

$2(8u)(-11v) = -176uv$, $2(8u)(-2w) = -32uw$, $2(-11v)(-2w) = 44vw$

= **$(8u - 11v - 2w)^2$**

🤔 Think and Reflect · 4.6 Factorisation Without Using Algebra Tiles

1 James: $(a - b)^2(a + b) = (a^2 - 2ab + b^2)(a + b)$

Reshma: $(a - b)^2(a + b) = (a - b)[(a - b)(a + b)] = (a - b)(a^2 - b^2)$

Who is correct and why?

✔ ANSWER

James: $(a^2 - 2ab + b^2)(a + b) = a^3 + a^2b - 2a^2b - 2ab^2 + ab^2 + b^3 = a^3 - a^2b - ab^2 + b^3$

Reshma: $(a - b)(a^2 - b^2) = a^3 - ab^2 - a^2b + b^3$

Both are correct; both give $a^3 - a^2b - ab^2 + b^3$. Reshma's way is shorter as it uses the identity $(a - b)(a + b) = a^2 - b^2$.

⭐ 4.7 Finding New Identities

? Predict what $(x + y)(x^2 - xy + y^2)$ will be.

✔ ANSWER

$= x^3 - x^2y + xy^2 + x^2y - xy^2 + y^3$

$= \mathbf{x^3 + y^3}$

Think and Reflect $x - y$ is a factor of $x^2 - y^2$ and $x^3 - y^3$. Is $x - y$ also a factor of $x^4 - y^4$? How about $x^5 - y^5$?

✔ ANSWER

$x^4 - y^4 = (x^2 - y^2)(x^2 + y^2) = \mathbf{(x - y)(x + y)(x^2 + y^2)}$, so yes.

$x^5 - y^5$ = **$(x - y)(x^4 + x^3y + x^2y^2 + xy^3 + y^4)$**, so yes, $x - y$ is a factor of $x^5 - y^5$ too.

(Check by multiplying: all middle terms cancel.)

🤔 Think and Reflect · 4.8 Simplifying Rational Expressions

1 **Simplify $(36s^2 - 12st + t^2)/(t^2 + 2ts - 48s^2) = (6s - t)^2/((___ + ___)(___ + ___))$**

✔ ANSWER

$t^2 + 2ts - 48s^2$: need two numbers with sum 2 and product -48 → 8 and -6.

$t^2 + 2ts - 48s^2$ = **$(t + 8s)(t - 6s)$**

$(6s - t)^2 = (t - 6s)^2$, so cancel one $(t - 6s)$:

= **$(t - 6s)/(t + 8s)$**

📝 Exercise Set 4.5

1 (i) **$(3p^2 - 3pq - 18q^2)/(p^2 + 3pq - 10q^2)$**

💡 SAMPLE ANSWER

Numerator = $3(p^2 - pq - 6q^2) = 3(p - 3q)(p + 2q)$

Denominator = $(p + 5q)(p - 2q)$

= **$3(p - 3q)(p + 2q) / ((p + 5q)(p - 2q))$**

There is no common factor, so this is the simplest form.

1 (ii) $(n^3 - 3n^2m + 3nm^2 - m^3)/(5m^2 - 10mn + 5n^2)$

✔ ANSWER

Numerator = $(n - m)^3$

Denominator = $5(m^2 - 2mn + n^2) = 5(n - m)^2$

= $\mathbf{(n - m)/5}$

1 (iii) $(w^3 - v^3 + x^3 + 3wvx)/(w^2 + v^2 + x^2 - 2wv - 2vx + 2wx)$

✔ ANSWER

Numerator = $w^3 + (-v)^3 + x^3 - 3(w)(-v)(x) = (w - v + x)(w^2 + v^2 + x^2 + wv + vx - wx)$

Denominator = $(w - v + x)^2$

= $\mathbf{(w^2 + v^2 + x^2 + wv + vx - wx)/(w - v + x)}$

1 (iv) $(4y^2 - 20yz + 25z^2)/(25z^2 - 4y^2)$

✔ ANSWER

Numerator = $(5z - 2y)^2$

Denominator = $(5z - 2y)(5z + 2y)$

= $\mathbf{(5z - 2y)/(5z + 2y)}$

1 (v) $(x^2 + x - 6)(x^2 - 7x + 12) / ((x^2 - 6x + 8)(x^2 - 9))$

✔ ANSWER

$x^2 + x - 6 = (x + 3)(x - 2)$; $x^2 - 7x + 12 = (x - 3)(x - 4)$

$x^2 - 6x + 8 = (x - 2)(x - 4)$; $x^2 - 9 = (x - 3)(x + 3)$

All factors cancel: = **1**

1 (vi) **$(p^4 - 16)/(p^2 - 4p + 4)$**

✔ ANSWER

$p^4 - 16 = (p^2 - 4)(p^2 + 4) = (p - 2)(p + 2)(p^2 + 4)$

$p^2 - 4p + 4 = (p - 2)^2$

= **$(p + 2)(p^2 + 4)/(p - 2)$**

Example 17 **Now draw Saira's rectangle ($x^2 + 8x + 15$) using the pieces.**

💡 SAMPLE ANSWER

Put the x × x square in a corner.

Place 5 x-strips along one side and 3 x-strips along the other side.

Fill the corner with the 15 unit squares in a 3 by 5 array.

The rectangle is **(x + 5) by (x + 3)**.

End-of-Chapter Exercises

1 (i) **$(-3x + 4)^2$**

✔ ANSWER

$(-3x)^2 + 2(-3x)(4) + 4^2$

= $9x^2 - 24x + 16$

1 (ii) $(2s + 7)(2s - 7)$

✔ ANSWER

$(2s)^2 - 7^2 = 4s^2 - 49$

1 (iii) $(p^2 + 1/2)(p^2 - 1/2)$

✔ ANSWER

$(p^2)^2 - (1/2)^2 = p^4 - 1/4$

1 (iv) $(2n + 7)(2n - 7)$

✔ ANSWER

$(2n)^2 - 7^2 = 4n^2 - 49$

1 (v) $(s - 2t)(s^2 + 2st + 4t^2)$

✔ ANSWER

Use $(x - y)(x^2 + xy + y^2) = x^3 - y^3$ with $x = s$, $y = 2t$:

$= s^3 - 8t^3$

1 (vi) $(1/(2r) - 4r)^2$

✔ ANSWER

$(1/(2r))^2 - 2(1/(2r))(4r) + (4r)^2$

$= 1/(4r^2) - 4 + 16r^2$

1 (vii) $(-3m + 4k - l)^2$

✔ ANSWER

$(-3m)^2 + (4k)^2 + (-l)^2 + 2(-3m)(4k) + 2(4k)(-l) + 2(-l)(-3m)$

$= \mathbf{9m^2 + 16k^2 + l^2 - 24mk - 8kl + 6lm}$

1 (viii) $(x - 1/3\ y)^3$

✔ ANSWER

$x^3 - 3x^2(1/3\ y) + 3x(1/3\ y)^2 - (1/3\ y)^3$

$= \mathbf{x^3 - x^2y + 1/3\ xy^2 - 1/27\ y^3}$

1 (ix) $(7/2\ k - 2/3\ m)^3$

✔ ANSWER

$(7/2\ k)^3 - 3(7/2\ k)^2(2/3\ m) + 3(7/2\ k)(2/3\ m)^2 - (2/3\ m)^3$

$= 343/8\ k^3 - 3 \times 49/4 \times 2/3\ k^2m + 3 \times 7/2 \times 4/9\ km^2 - 8/27\ m^3$

$= \mathbf{343/8\ k^3 - 49/2\ k^2m + 14/3\ km^2 - 8/27\ m^3}$

2 (i) 17×21

✔ ANSWER

$(19 - 2)(19 + 2) = 19^2 - 2^2 = 361 - 4 = \mathbf{357}$

2 (ii) 104×96

✔ ANSWER

$(100 + 4)(100 - 4) = 10000 - 16 = \mathbf{9984}$

2 (iii) **24×16**

✔ ANSWER

$(20 + 4)(20 - 4) = 400 - 16 = \mathbf{384}$

2 (iv) **147^3**

✔ ANSWER

$(150 - 3)^3 = 150^3 - 3(150)^2(3) + 3(150)(3)^2 - 3^3$

$= 3375000 - 202500 + 4050 - 27 = \mathbf{3176523}$

2 (v) **199^3**

✔ ANSWER

$(200 - 1)^3 = 8000000 - 3(40000)(1) + 3(200)(1) - 1$

$= 8000000 - 120000 + 600 - 1 = \mathbf{7880599}$

2 (vi) **127^3**

✔ ANSWER

$(120 + 7)^3 = 120^3 + 3(120)^2(7) + 3(120)(7)^2 + 7^3$

$= 1728000 + 302400 + 17640 + 343 =$ **2048383**

2 (vii) $(-107)^3$

✔ ANSWER

$(-107)^3 = -(100 + 7)^3 = -(1000000 + 210000 + 14700 + 343)$

$= \mathbf{-1225043}$

2 (viii) $(-299)^3$

✔ ANSWER

$(-299)^3 = -(300 - 1)^3 = -(27000000 - 270000 + 900 - 1)$

$= \mathbf{-26730899}$

3 (i) $4y^2 + 1 + 1/(16y^2)$

✔ ANSWER

$= (2y)^2 + 2(2y)(1/(4y)) + (1/(4y))^2$

$= \mathbf{(2y + 1/(4y))^2}$

3 (ii) $9m^2 - 1/(25n^2)$

✔ ANSWER

$= (3m)^2 - (1/(5n))^2$

$= \mathbf{(3m + 1/(5n))(3m - 1/(5n))}$

3 (iii) $27b^3 - 1/(64b^3)$

✔ ANSWER

= $(3b)^3 - (1/(4b))^3$. Use $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$:

= **$(3b - 1/(4b))(9b^2 + 3/4 + 1/(16b^2))$**

3 (iv) **$x^2 + 5x/6 + 1/6$**

✔ ANSWER

Need a + b = 5/6, ab = 1/6 → 1/2 and 1/3.

= **$(x + 1/2)(x + 1/3)$**

3 (v) **$27u^3 - 1/125 - 27u^2/5 + 9u/25$**

✔ ANSWER

= $(3u)^3 - 3(3u)^2(1/5) + 3(3u)(1/5)^2 - (1/5)^3$

= **$(3u - 1/5)^3$**

3 (vi) **$64y^3 + z^3/125$**

✔ ANSWER

= $(4y)^3 + (z/5)^3$. Use $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$:

= **$(4y + z/5)(16y^2 - 4yz/5 + z^2/25)$**

3 (vii) **$p^3 + 27q^3 + r^3 - 9pqr$**

✔ ANSWER

= $p^3 + (3q)^3 + r^3 - 3(p)(3q)(r)$

= **$(p + 3q + r)(p^2 + 9q^2 + r^2 - 3pq - 3qr - pr)$**

3 (viii) $9m^2 - 12m + 4$

✔ ANSWER

$= (3m)^2 - 2(3m)(2) + 2^2 =$ **$(3m - 2)^2$**

3 (ix) $9x^3 - 8/3\ y^3 + z^3/3 + 6xyz$

✔ ANSWER

Take 1/3 out: $= 1/3\ (27x^3 - 8y^3 + z^3 + 18xyz)$

$= 1/3\ [(3x)^3 + (-2y)^3 + z^3 - 3(3x)(-2y)(z)]$

$=$ **$1/3\ (3x - 2y + z)(9x^2 + 4y^2 + z^2 + 6xy + 2yz - 3xz)$**

3 (x) $4x^2 + 9y^2 + 36z^2 + 12xz + 36yz + 24xy$

💡 SAMPLE ANSWER

Squares: $(2x)^2$, $(3y)^2$, $(6z)^2$. For $(2x + 3y + 6z)^2$ the middle terms are $12xy + 36yz + 24xz$.

In the book the xy and xz terms appear swapped (24xy, 12xz); then it is not a perfect square. It is most likely a printing mistake.

With $12xy + 24xz$: **$(2x + 3y + 6z)^2$**

3 (xi) $27u^3 - 1/216 - 9u^2/2 + u/4$

✔ ANSWER

$= (3u)^3 - 3(3u)^2(1/6) + 3(3u)(1/6)^2 - (1/6)^3$

$=$ **$(3u - 1/6)^3$**

4 (i) **$(4x^2 + 4x + 1)/(4x^2 - 1)$**

✔ ANSWER

$= (2x + 1)^2 / ((2x + 1)(2x - 1))$

$=$ **$(2x + 1)/(2x - 1)$**

4 (ii) **$9(3a^3 - 24b^3)/(9a^2 - 36b^2)$**

✔ ANSWER

Numerator $= 27(a^3 - 8b^3) = 27(a - 2b)(a^2 + 2ab + 4b^2)$

Denominator $= 9(a^2 - 4b^2) = 9(a - 2b)(a + 2b)$

$=$ **$3(a^2 + 2ab + 4b^2)/(a + 2b)$**

4 (iii) **$(s^3 + 125t^3)/(s^2 - 2st - 35t^2)$**

✔ ANSWER

Numerator $= (s + 5t)(s^2 - 5st + 25t^2)$

Denominator $= (s - 7t)(s + 5t)$

$=$ **$(s^2 - 5st + 25t^2)/(s - 7t)$**

5 (i) **Rectangle of area $25a^2 - 30ab + 9b^2$**

✔ ANSWER

$= (5a - 3b)^2$

Length = 5a − 3b, breadth = 5a − 3b (it is a square).

5 (ii) **Rectangle of area $36s^2 - 49t^2$**

✔ ANSWER

$= (6s + 7t)(6s - 7t)$

Length = 6s + 7t, breadth = 6s − 7t

6 (i) **Cuboid of volume $6a^2 - 24b^2$**

✔ ANSWER

$= 6(a^2 - 4b^2) = 6(a + 2b)(a - 2b)$

Length = a + 2b, breadth = a − 2b, height = 6 (other choices like 2, 3(a + 2b), a − 2b also work).

6 (ii) **Cuboid of volume $3ps^2 - 15ps + 12p$**

✔ ANSWER

$= 3p(s^2 - 5s + 4) = 3p(s - 1)(s - 4)$

Length = s − 1 (or s − 4), breadth = s − 4 (or s − 1), height = 3p

7 **Square playground of side 40 m; a path of width s m is made around it. Find the area of the path.**

💡 SAMPLE ANSWER

Outer square side = 40 + 2s

Area of path = $(40 + 2s)^2 - 40^2 = 1600 + 160s + 4s^2 - 1600$

$= \mathbf{4s^2 + 160s = 4s(s + 40)}$ **sq. m**

(If the path is inside the playground: $40^2 - (40 - 2s)^2 = 160s - 4s^2$.)

8 **If a number plus its reciprocal equals 10/3, find the number.**

✔ ANSWER

$x + 1/x = 10/3 \rightarrow 3x^2 + 3 = 10x \rightarrow 3x^2 - 10x + 3 = 0$

$3x^2 - 9x - x + 3 = 3x(x - 3) - 1(x - 3) = (3x - 1)(x - 3) = 0$

x = 3 or x = 1/3

9 **A rectangular pool has area $2x^2 + 7x + 3$ square hastas and width $2x + 1$ hastas. Find its length.**

✔ ANSWER

$2x^2 + 7x + 3 = 2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$

Length = x + 3 hastas

10* **If both $x - 2$ and $x - 1/2$ are factors of $px^2 + 5x + r$, show that $p = r$.**

✔ ANSWER

$x - 2$ is a factor, so the value at $x = 2$ is 0: $4p + 10 + r = 0$ ... (1)

$x - 1/2$ is a factor, so at $x = 1/2$: $p/4 + 5/2 + r = 0 \rightarrow p + 10 + 4r = 0$ ... (2)

(1) − (2): 3p − 3r = 0

So p = r. (In fact p = r = −2.)

11* **If a + b + c = 5 and ab + bc + ca = 10, prove that $a^3 + b^3 + c^3 - 3abc = -25$.**

✔ ANSWER

$a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = 25 - 20 = 5$

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$= 5 \times (5 - 10) = 5 \times (-5) =$ **−25**. Proved.

12* **By factoring, check that $n^3 - n$ is always divisible by 6 for all natural numbers n. Give reasons.**

✔ ANSWER

$n^3 - n = n(n^2 - 1) =$ **(n − 1) n (n + 1)**

This is a product of 3 consecutive whole numbers.

Among any 2 consecutive numbers one is even → divisible by 2.

Among any 3 consecutive numbers one is a multiple of 3 → divisible by 3.

So the product is divisible by 2 × 3 = **6**. (For n = 1 it is 0, also divisible.)

13* (i) **Find $x^3 + y^3 - 12xy + 64$, when $x + y = -4$**

✔ ANSWER

Write it as $x^3 + y^3 + 4^3 - 3(x)(y)(4)$.

$= (x + y + 4)(x^2 + y^2 + 16 - xy - 4y - 4x)$

$x + y + 4 = -4 + 4 = 0$, so the value = **0**

13* (ii) **Find $x^3 - 8y^3 - 36xy - 216$, when $x = 2y + 6$**

✔ ANSWER

Write it as $x^3 + (-2y)^3 + (-6)^3 - 3(x)(-2y)(-6)$.

$= (x - 2y - 6)(\ldots)$

$x - 2y - 6 = 0$, so the value = **0**

🎉 **Chapter 4 done! Revise the answers once more tomorrow.**